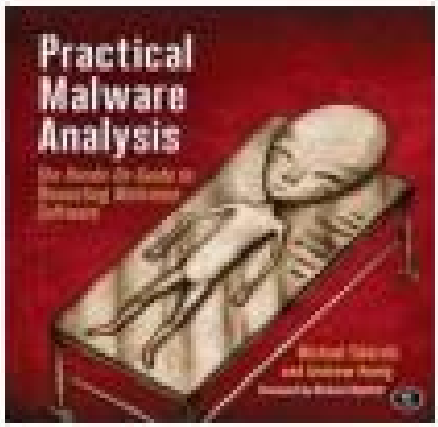
PMA Lab 1.1 Writeup:





https://www.linkedin.com/in/talha-yaqoob

Lab 1:

1. **Upload the files to http://www.VirusTotal.com/ and view the reports. Does either file match any existing antivirus signatures**



yep it does.

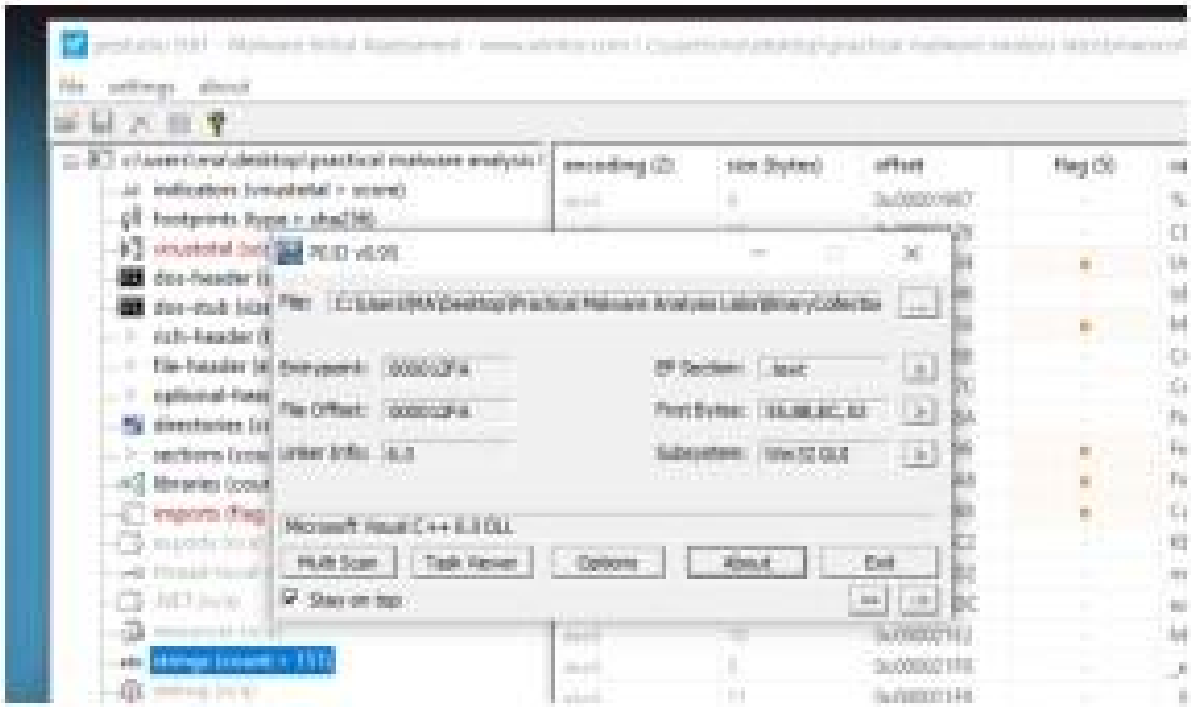
2.When were these files compiled?



NOTE:

This time and date can easily be altered.

3. Are there any indications that either of these files is packed or obfuscated? If so, what are these indicators?



property	value
section	section[0]
name	.text
section > sha256	44AB4D055ABE09F315F2172...
entropy	4.451
file > ratio (75.00%)	25.00 %
raw-address (begin)	0x00001000
raw-address (end)	0x00002000
raw-size (12288 bytes)	0x00001000 (4096 bytes)
virtual-address (begin)	0x00001000
virtual-address (end)	0x00001970
virtual-size (3358 bytes)	0x00000970 (2416 bytes)

Both the raw address and the virtual address have the same size so No there is packing and obfuscation



more than enough imports that makes the file and dll legit

4. Do any imports hint at what this malware does? If so, which imports are they?



And kernel32.dll

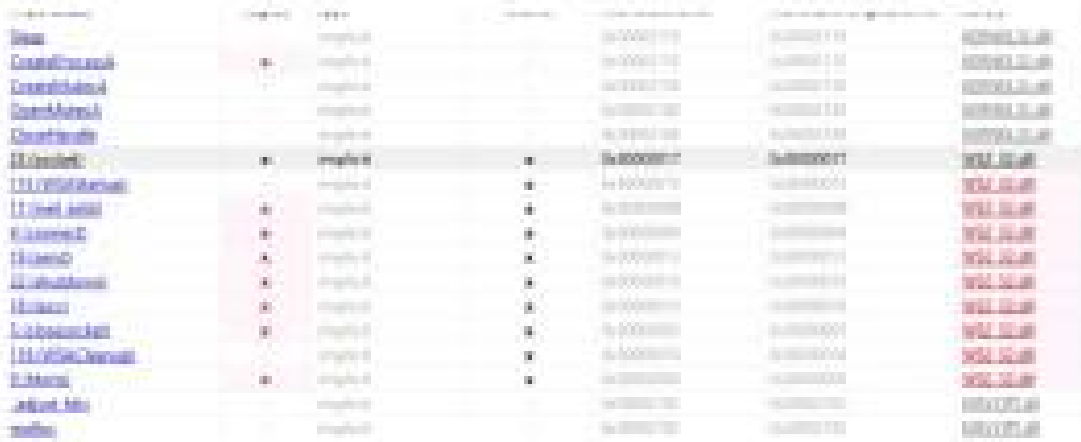
6. What network-based indicators could be used to find this malware on infected machines

string > utf-pattern	127.26.152.13

we find this specific ip address which it may look that it is a loop back address since it is starting from 127 but it can be a connection to an outside world

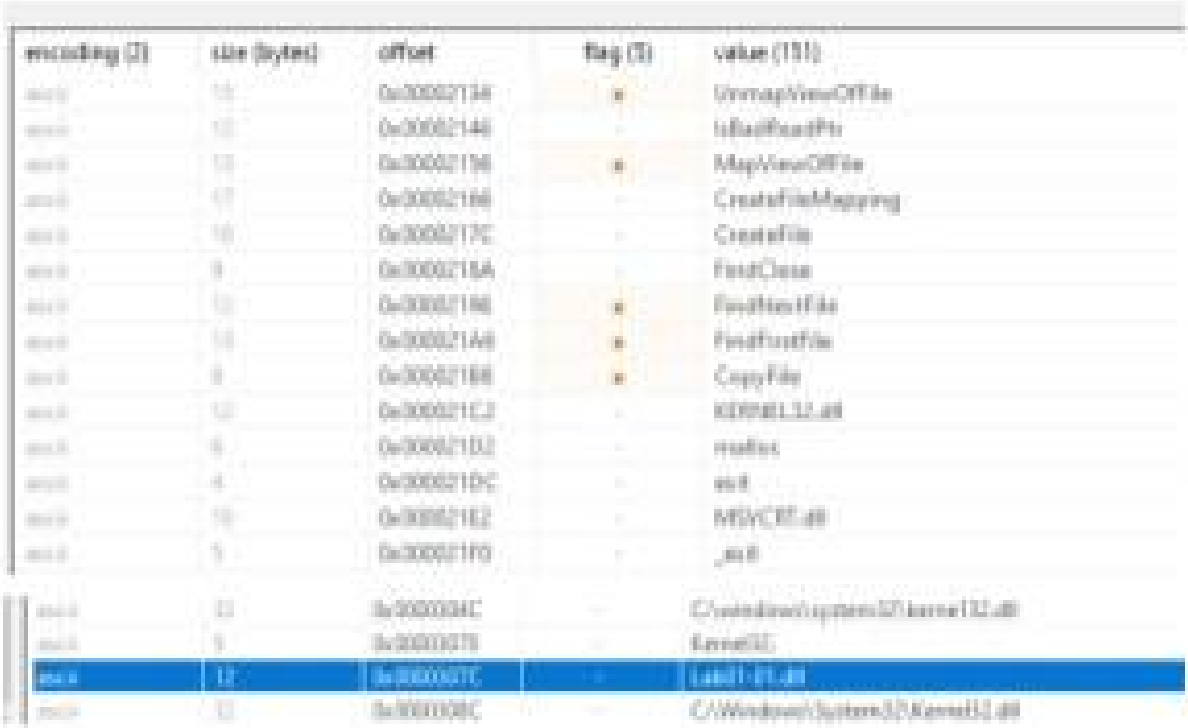


Okay this is clear indication that the dll has a functionality of establishing a communication to the outside world



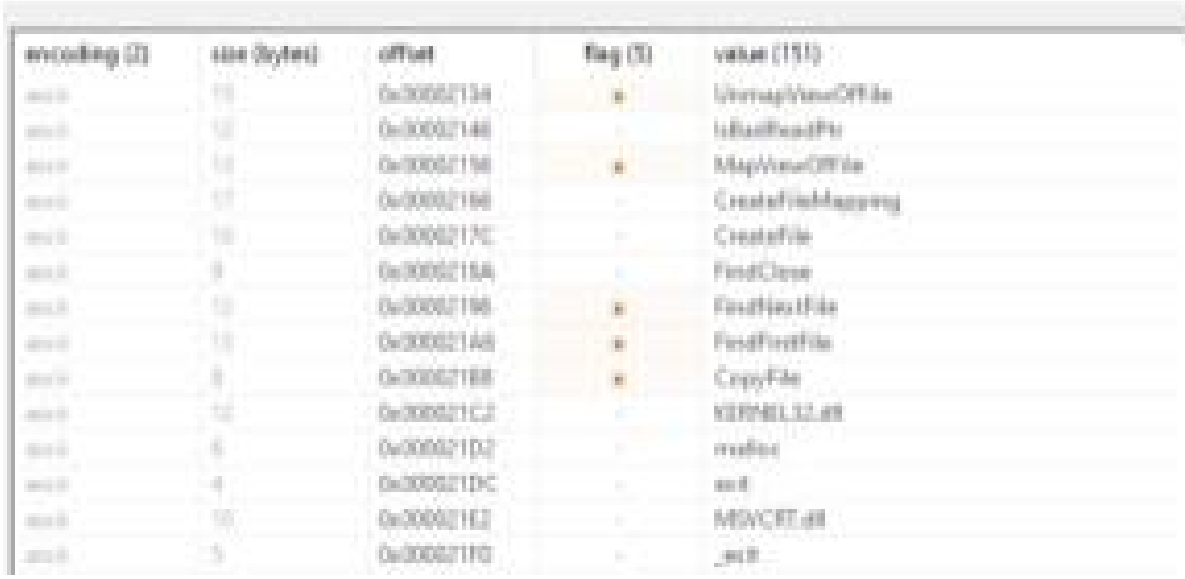
- **WS2.dll** (or WS2_32.dll) is the Windows DLL that handles **networking** (sockets, TCP, UDP, IP... all the internet magic).
- Inside that DLL, every function has an **internal ID number**.
- **23** is the internal ID for the **socket()** function.

"23 socket WS2.dll" = the socket creation function inside Windows' networking DLL.
this is where it starts creating network connections



encoding (2)	size (bytes)	offset	flag (5)	value (151)
ascii	13	0x30002134	x	UnmapViewOfFile
ascii	12	0x30002146	-	IsBadReadPtr
ascii	13	0x30002156	x	MapViewOfFile
ascii	17	0x30002166	-	CreateFileMapping
ascii	10	0x3000217C	-	CreateFile
ascii	9	0x30002186	-	FindClose
ascii	12	0x30002190	x	FindNextFile
ascii	13	0x300021A0	x	FindFirstFile
ascii	8	0x300021B0	x	CopyFile
ascii	12	0x300021C2	-	KERNEL32.dll
ascii	6	0x300021D2	-	malloc
ascii	4	0x300021DC	-	exit
ascii	10	0x300021E2	-	MSVCRT.dll
ascii	5	0x300021F0	-	_exit
ascii	13	0x3000304C	-	C:\windows\system32\kernel32.dll
ascii	9	0x3000307B	-	Kernel32
ascii	12	0x3000307C	-	Lab01-01.dll
ascii	13	0x3000308C	-	C:\Windows\System32\Kernel32.dll

5. Are there any other files or host-based indicators that you could look for on infected systems?



encoding (2)	size (bytes)	offset	flag (5)	value (151)
ascii	13	0x30002134	x	UnmapViewOfFile
ascii	12	0x30002146	-	IsBadReadPtr
ascii	13	0x30002156	x	MapViewOfFile
ascii	17	0x30002166	-	CreateFileMapping
ascii	10	0x3000217C	-	CreateFile
ascii	9	0x30002186	-	FindClose
ascii	12	0x30002190	x	FindNextFile
ascii	13	0x300021A0	x	FindFirstFile
ascii	8	0x300021B0	x	CopyFile
ascii	12	0x300021C2	-	KERNEL32.dll
ascii	6	0x300021D2	-	malloc
ascii	4	0x300021DC	-	exit
ascii	10	0x300021E2	-	MSVCRT.dll
ascii	5	0x300021F0	-	_exit

It find the files in the C drive

Yes , as virus total states " It drops a copy of a DLL named "Lab01-01.dll" as "C:\Windows\system32\kerne132.dll" . It then recursively traverses the file system starting from "C:*", finds all .exe files, and patches their import tables to replace imports from "kernel32.dll" with the malicious "kerne132.dll" which is malicious dll that connect to the outer ip address so indication of such activity.

7. What would you guess is the purpose of these files?

- As may it seem, the file replaces kernel32.dll with kerne132.dll which is a malicious dll and when executed the program will import malicious function
- The file then iterates through the whole C drive



- The dll calls CreateMutexA is used to create a new mutex object. Mutexs are often used by malware to prevent the reinfection of a system with the same or different malware variant.
- It imports the lab1.dll



- The exe installs the dll in the C: windows\system 32 its C2 server.
- • **The Action:** The `.exe` runs, finds this `.dll`, and copies it into `C:\Windows\System32\` renaming it to `kerne132.dll` to establish **persistence**.
- • **The Goal:** It attempts to hide in plain sight so that it runs or sits undetected in a critical system folder. It acts as a backdoor.
- Then the dll connects to the C2

This is what virus total concludes:

